Case Study: Immersive Gallery Experience

Role: Designer (UX/UI & Prototyping) & Developer - Jennifer Lindsey

Figma Prototype | Github repo | Live Gallery App

The Problem: Online museum archives are often data graveyards. Learners feel overwhelmed by thousands of thumbnails and lack the close-up intimacy required to appreciate art techniques like Chiaroscuro.

Learning Goals:

1. Describe how artists use contrast of light and dark colors to direct viewers attention.
2. Point to visible evidence of artist's technique such as brushstrokes or paint thickness in detail of painting.
3. Use specialized vocabulary to explain what they see. (chiaroscuro, tenebrism).

Key Terms:
Chiaroscuro- italian term meaning 'light-dark'. In art the term refers to the tonal contrasts, light and dark areas within a painting, creating light and shadow.

Tenebrism - a term derived from the Italian 'tenebroso' which means darkened and obscuring. It is used to describe a certain type of painting in which significant details such as faces and hands are illuminated by highlights which are contrasted with a predominantly dark or black background. The late paintings of Caravaggio and those of many of his followers are often described as tenebrist.

1. Features

From my years as a museum educator and art history professor, I watched how people "lean in" and "lean out" to get a closer look at details. Digital archives are usually static and do not always support this.

Feature	What the learner does	Learning mechanism (hypothesis)
Light Study sliders	Adjust simulated lighting to explore chiaroscuro effects interactively.	Active manipulation: changing one variable at a time lets learners see how light directs attention, instead of being told.
Click to examine details	Moves in on a detail like the beard brushwork with click to toggle 2.5× magnification	Revealing brushwork and materiality hidden at standard scale. Sustained attention: seeing material evidence of technique supports the "slow looking" habit and gives them something concrete to describe.
Glossary on demand	Opens definitions when learner clicks on Exhibition Info button	on-click support: vocabulary available when it's needed, so the term and the experience connect.
Conservator mode	Switches to a high-contrast analytical view	Multiple ways to engage: supports different learners, including those with low vision.
Curated filtering	Sees works filtered by keyword, period, date range (1580-1750), and with images	Trustworthy content: removes distractions and builds confidence in what's being taught.

2. Information Architecture: Curated Filtering

I architected a custom composable to interface with the Metropolitan Museum of Art's API. The logic programmatically ensures that every API request is tethered to the 'Baroque' keyword. This prevents the noisy or off-period results often found in open-source museum APIs, allowing me to transform a raw data stream into a curated art-historical exhibition.

Filtering Process to maintain curated exhibition:
Pass 1: API Fetch.
Pass 2: Period and date filter (Filtering for 1580–1750).
Final Pass 3: Image check (each record must include an image).

3. Prototyping for Validation

To test if the "Light Study" feature was intuitive, I built a functional prototype. Static Figma mocks couldn't capture the physics of light moving across a canvas.

- **Learner Insight:** Informal early testing with 5 learners showed that learners didn't all understand the "chiaroscuro" concept without a visual trigger. I added a **Glossary Dialog** when learner clicks **Exhibition Info** button that provides context, so learners can can look up terms when needed closing the loop between *seeing* and *learning*.
    - **Learner Problem:** Learners may find term "Baroque" art intimidating or technically confusing.
    - **The Solution:** a clickable button with high-contrast as an entry point to a glossary.
    - **Question:** Not yet tested: whether the glossary improves learners' use of the vocabulary. It is available for defining terms.

4. Evaluation Questions

Learning goal	Question	What I'd log or collect
1. Light directs attention	Do learners who use the Light Study sliders describe light and contrast more specifically?	Slider use (which slider, how much, how long) and a short written response: "Where does your eye go first, and what makes it go there?"
2. Evidence of technique	Do learners who zoom in point to specific details like brushwork?	Zoom toggles, which regions were examined, time spent zoomed, and a written response naming one detail and what it shows.
3. Vocabulary	Does opening the glossary lead to using the terms in descriptions?	Glossary opens, and whether terms like chiaroscuro appear in the written response.
Access	Do learners choose Conservator mode, and does it help them read the metadata?	Mode switches and when they happen, plus a short preference question. Also test keyboard-only use and a screen reader pass.

5. What I Learned

- Interaction is a Form of Scholarship:
  I hypothesize that digital features like the Conservator Light Mode or Click to see Details (Mouse-Tracking Zoom) are not just nice-to-have features; they are pedagogical tools. By allowing the zoom to follow the learner's natural attention, I realized that the transition is where the learning happens. Design may help a learner begin to see like a conservator.
- Guardrails Preserve Integrity
  Working with the Met's Open API taught me that raw data is rarely exhibition-ready. I was able to bridge the gap between Software Engineering and Digital Stewardship by architecting a custom logic layer. Building the useMetArtAPI composable taught me that expert-defined filtering rules means using code to enforce curatorial boundaries and keep the exhibition focused on the period, ensuring that the digital archive remains as academically rigorous as a physical museum.